

YORK RIVER EAST RANGE LIGHT, VA, USA

WMO: 997708

Lat: 37.251N

Lon: 76.342W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.6	(c) -3.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 15.6	(k) 1.0	(l) 14.6	(m) 1.4	(n) 7.5	(o) 0	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.5	(c) 31.6	(d) N/A	(e) 30.5	(f) N/A	(g) 29.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.3	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.7	(b) 12.6	(c) 11.4	(d) DB	(e) -7.5	(f) 34.1	(g) 2.7	(h) 1.3	(i) -9.5	(j) 35.1	(k) -11.1	(l) 35.8	(m) -12.6	(n) 36.6	(o) -14.6	(p) 37.5	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.2	5.3	8.5	14.7	19.9	24.5	26.5	25.9	23.0	17.6	11.2	8.1
	DBStd	8.63	4.86	4.47	4.48	4.04	3.37	2.59	1.95	1.79	2.47	3.67	3.65	4.28
	HDD10.0	538	187	143	84	7	0	0	0	0	0	2	29	87
	HDD18.3	1840	437	365	305	123	23	0	0	0	1	56	214	317
	CDD10.0	2670	9	11	38	147	306	435	511	492	389	238	66	29
	CDD18.3	931	0	0	1	13	71	186	252	233	141	34	1	0
	CDH23.3	6568	0	0	2	36	236	1281	2344	1946	668	56	0	0
	CDH26.7	1403	0	0	0	3	34	277	620	393	75	1	0	0
(14) Wind	WSAvg	6.0	6.3	6.5	6.6	6.5	6.0	5.5	5.2	5.4	5.8	6.3	6.3	5.9
(15) Precipitation	PrecAvg	1177	84	77	99	80	89	98	138	130	127	95	73	89
	PrecMax	1655	155	246	239	143	141	227	234	287	429	227	252	190
	PrecMin	798	36	30	18	17	27	36	28	31	23	3	3	36
	PrecStd	203	33	45	42	31	28	42	55	72	90	53	54	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.5	22.1	26.4	29.5	32.9	33.9	32.4	30.2	26.5	20.7	20.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	16.1	16.6	19.0	24.2	27.5	30.6	31.8	31.0	28.6	24.8	18.6	18.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	13.7	14.5	17.2	22.3	25.8	29.5	30.7	29.8	27.4	23.7	17.4	16.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	11.1	11.8	15.1	20.4	24.4	28.2	29.7	28.8	26.5	22.5	16.2	14.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.0	5.8	5.9	5.0	4.7	4.5	4.0	3.7	4.7	4.9	5.4
		MCDBR	9.3	9.6	9.4	8.5	6.8	6.5	6.3	5.6	5.2	5.7	6.1	8.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.306	0.317	0.344	0.393	0.435	0.484	0.515	0.510	0.411	0.353	0.322	0.310
	taud		2.546	2.510	2.466	2.339	2.253	2.165	2.093	2.093	2.359	2.523	2.559	2.568
	Ebn at Noon		889	921	921	888	852	807	778	772	840	868	863	860
	Edh at Noon		77	90	103	124	137	150	160	156	114	88	75	71
(44) All-Sky Solar Radiation	RadAvg		2.31	3.08	4.11	5.22	5.90	6.37	6.08	5.49	4.45	3.51	2.62	2.05
	RadStd		0.19	0.33	0.33	0.38	0.53	0.43	0.29	0.29	0.38	0.40	0.22	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	+103

Nomenclature: See separate page